



△ IC 5148

A ghostly shell of gas cast off by a dying star and resembling a car tire gives this planetary nebula its popular name of the Spare Tire Nebula. The gas shell is a couple of light-years across and is speeding away from a white dwarf (the bright white object in the center of the planetary nebula), which is the remnant of the original star.

KEY DATA

Size ranking 45

Brightest stars Alnair (α)
1.7, Beta (β) 2.0–2.3

Genitive Gruis

Abbreviation Gru

Highest in sky at 10pm
September–October

Fully visible 33°N–90°S



CHART 3

MAIN STARS

Alnair Alpha (α) Gruis
Blue-white subgiant

☼ 1.7 ↔ 101 light-years

Beta (β) Gruis
Variable red giant

☼ 2.0–2.3 ↔ 177 light-years

Gamma (γ) Gruis
Blue-white subgiant

☼ 3.0 ↔ 210 light-years

DEEP-SKY OBJECTS

NGC 7424
Intermediate spiral galaxy

IC 5148 (Spare Tire Nebula)
Planetary nebula

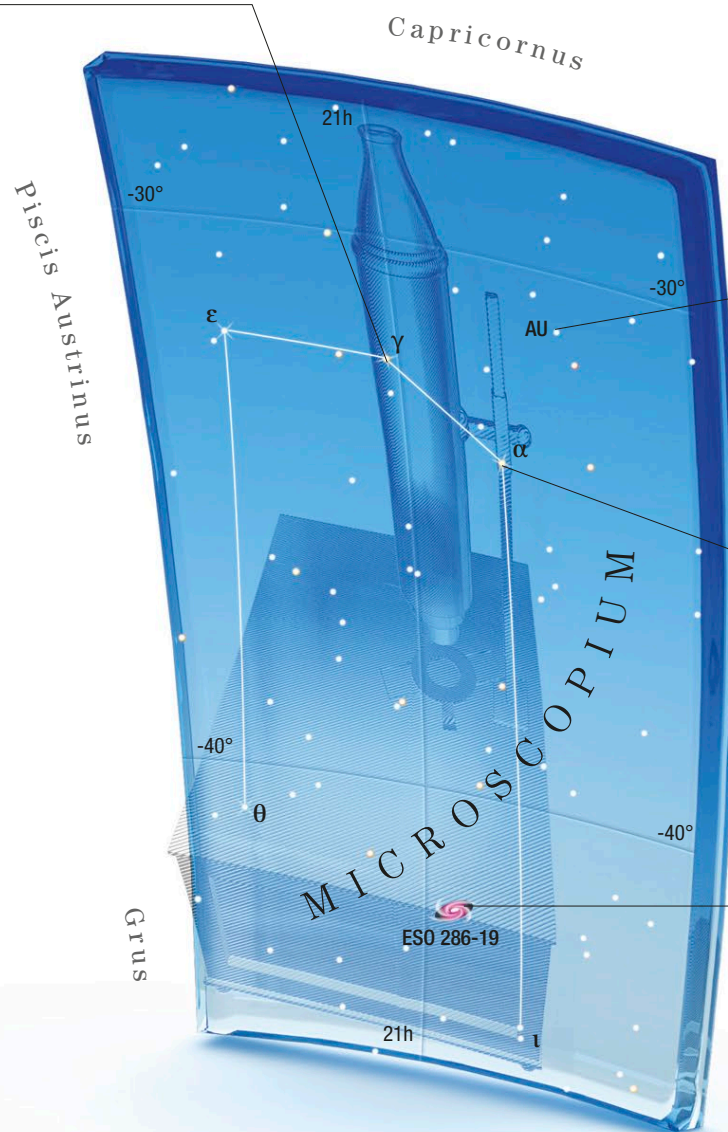
MICROSCOPIUM

THE MICROSCOPE

A SMALL AND FAINT SOUTHERN CONSTELLATION ADDED TO THE SKY IN THE MID-18TH CENTURY, MICROSCOPIUM IS A ROUGHLY RECTANGULAR PATTERN OF INDISTINCT STARS.

Microscopium is one of the 14 constellations introduced to the sky by the French astronomer Nicholas Louis de Lacaille. South of Capricornus, it is located between the more prominent constellations of Piscis Austrinus and Sagittarius. Microscopium is an almost featureless constellation with no bright stars and no deep-sky objects except for galaxies too faint for amateur telescopes. Theta is the brightest of several variable stars but its variations are difficult to see, only differing by 0.1 magnitude.

Gamma (γ) Microscopii
Yellow giant about 10 times the size of the Sun and 2.5 times its mass; magnitude 4.7



KEY DATA

Size ranking 66

Brightest stars Gamma (γ)
4.7, Epsilon (ε) 4.7

Genitive Microscopii

Abbreviation Mic

Highest in sky at 10pm
August–September

Fully visible 62°N–90°S



CHART 4

MAIN STARS

Alpha (α) Microscopii
Yellow giant

☼ 4.9 ↔ 380 light-years

Gamma (γ) Microscopii
Yellow giant

☼ 4.7 ↔ 230 light-years

Epsilon (ε) Microscopii
White main-sequence star

☼ 4.7 ↔ 180 light-years

DEEP-SKY OBJECTS

ESO 286-19

Two colliding galaxies

Debris disk around AU Microscopii

Dusty material in orbit around a young star

AU Microscopii
Faint red dwarf, about 32 light-years away and surrounded by a ring of dusty, potentially planet-forming material

Alpha (α) Microscopii
Yellow giant about 16 times the Sun's diameter and 160 times its luminosity. It forms an optical double with a magnitude 10 star

ESO 286-19
An unusual object lying about 600 light-years away and consisting of two previously disk-shaped galaxies in the midst of an ongoing collision